GlucoTwin

A two-hour warning, and a what-if sandbox, for type 2 diabetes.

six seven

India does not have a diabetes problem. It has a daily one.

- 101 million people with diabetes in India. 136 million with prediabetes.
- Most of them see a doctor two or three times a year.
- The disease acts every day, between those visits.

Anjana et al., Lancet Diabetes & Endocrinology, 2023. ICMR-INDIAB national study.

The dangerous hours are the ones nobody sees.

- Glucose above 180 mg/dL for 30 minutes does damage quietly.
- The worst window is usually after the evening meal.
- A 09:00 fasting reading cannot show it.
- The next HbA1c test arrives three months later.

So we rebuilt the patient as a twin.

- Static record: labs, medications, genotype, history.
- Dynamic stream: 14 days of CGM, heart rate, HRV, steps, sleep.
- Output: two-hour event risk, with a calibrated probability.
- Output: a what-if sandbox that runs on this patient's physiology.

1,500 synthetic patients. 14 days each.

6.05 M

glucose readings

5 min

sampling interval

55

model features

1 CPU

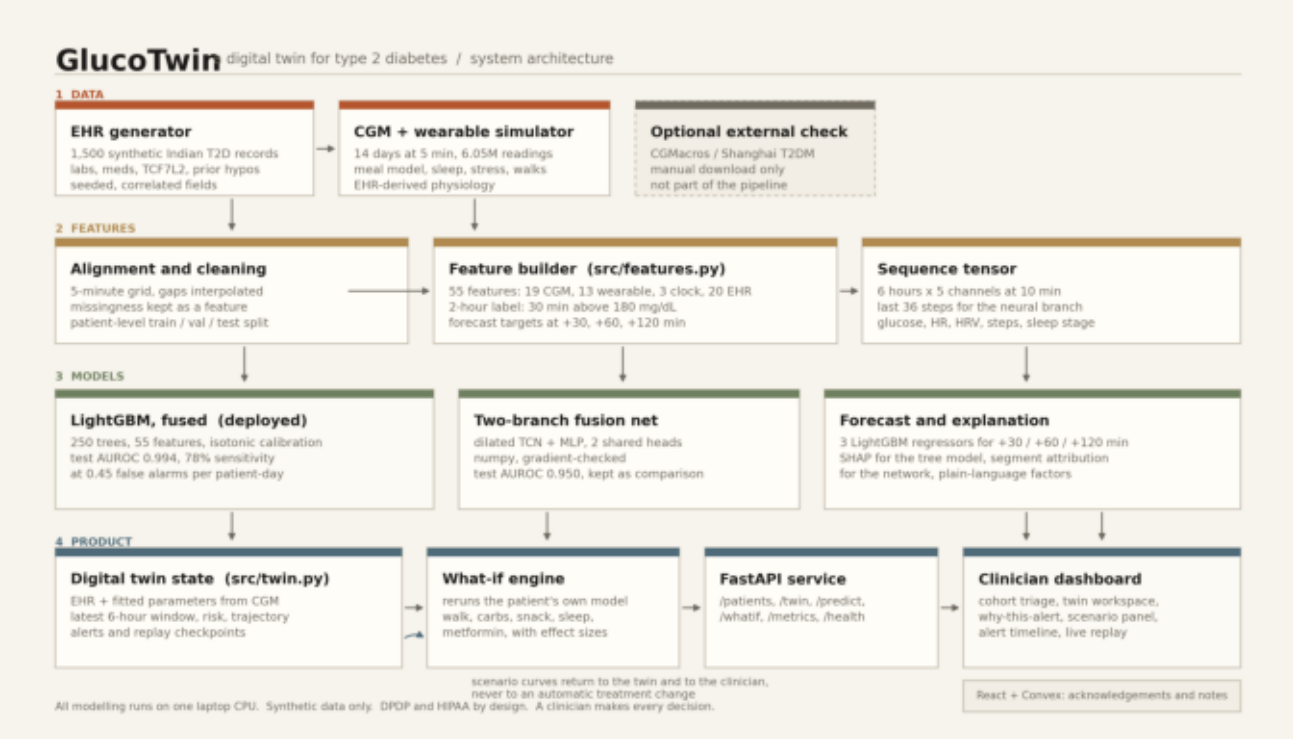
to train and run

Seed 42. The whole cohort is regenerated by one command.

Synthetic is a design choice, not a shortcut.

- No real patient data, so no consent or re-identification risk.
- The record drives the physiology, so the two streams are genuinely linked.
- Fasting glucose sets the baseline. BMI, duration and genotype set sensitivity.
- The cost: results do not transfer to real patients without validation.

One pipeline, from a synthetic record to a clinical screen.



Data generation, features, models, twin, what-if, API, dashboard. External validation is optional and off to the side.

The label is a real clinical event.

1 if glucose stays above 180 mg/dL for 30 minutes inside the next 2 hours

- 41.1% of test windows are positive.
- The cohort is deliberately uncontrolled, so events are not rare.
- Split by patient: 900 train, 300 validation, 300 test.

55 features. Four blocks. One fused vector.

- 19 CGM: level, means, variability, slopes, time above range, gaps.
- 13 wearable: heart rate, HRV, steps, sleep and deep sleep.
- 3 clock: time of day and night flag.
- 20 record: age, BMI, waist, HbA1c, eGFR, lipids, medications, TCF7L2.

Plus a 6-hour, 5-channel tensor for the neural branch.

Five models. The tree won.

Model	Input	AUROC	Sensitivity at 0.5 FA/day
Persistence	current glucose	0.797	6%
LightGBM, CGM only	15 CGM features	0.984	70%
LightGBM, EHR only	20 record fields	0.791	5%
LightGBM, fused	**all 55 features**	**0.994**	**78%**
Fusion net	6 h x 5 channels + record	0.950	24%

Test patients only. Calibrated probabilities. Threshold fixed on validation.

The network was supposed to win. It did not.

- Two-branch TCN + MLP, written in numpy, gradient-checked.
- AUROC 0.9496, against 0.9941 for the tree model.
- Two epochs on one CPU core is not enough to beat 250 boosted trees.
- So the tree is deployed, and the network stays in the ablation.

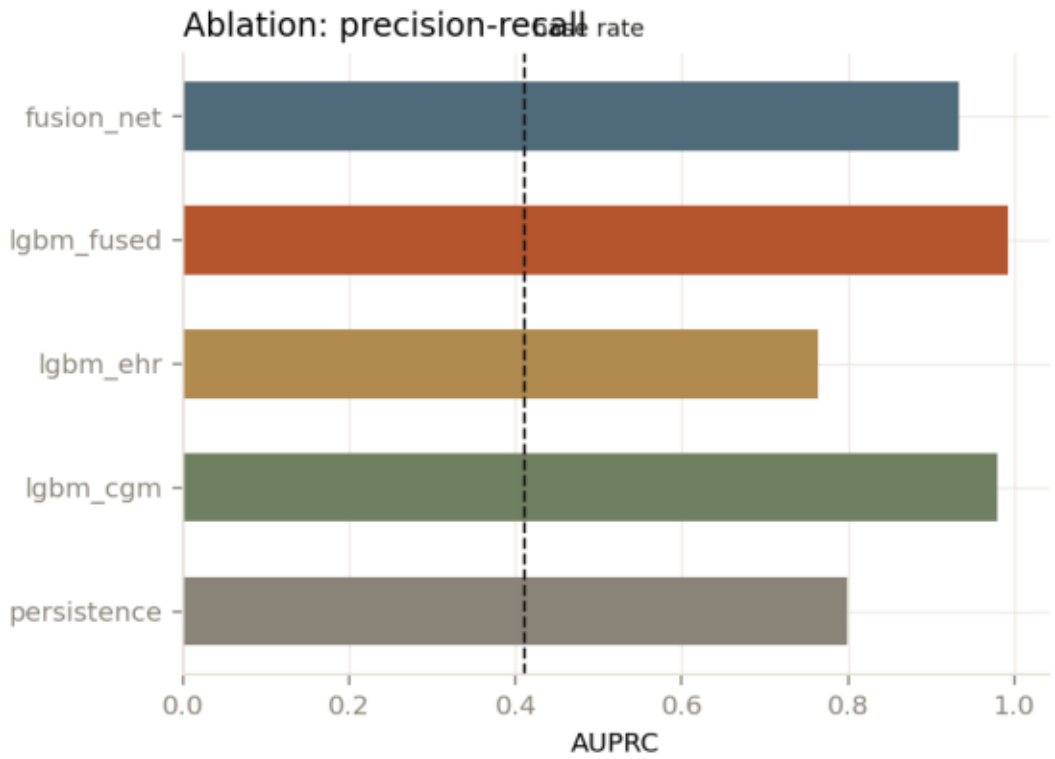
This is the comparison the brief asked for, reported as it is.

Results on 300 patients the model never saw.

0.994	0.992	78%	53 min
AUROC	AUPRC	sensitivity	mean warning

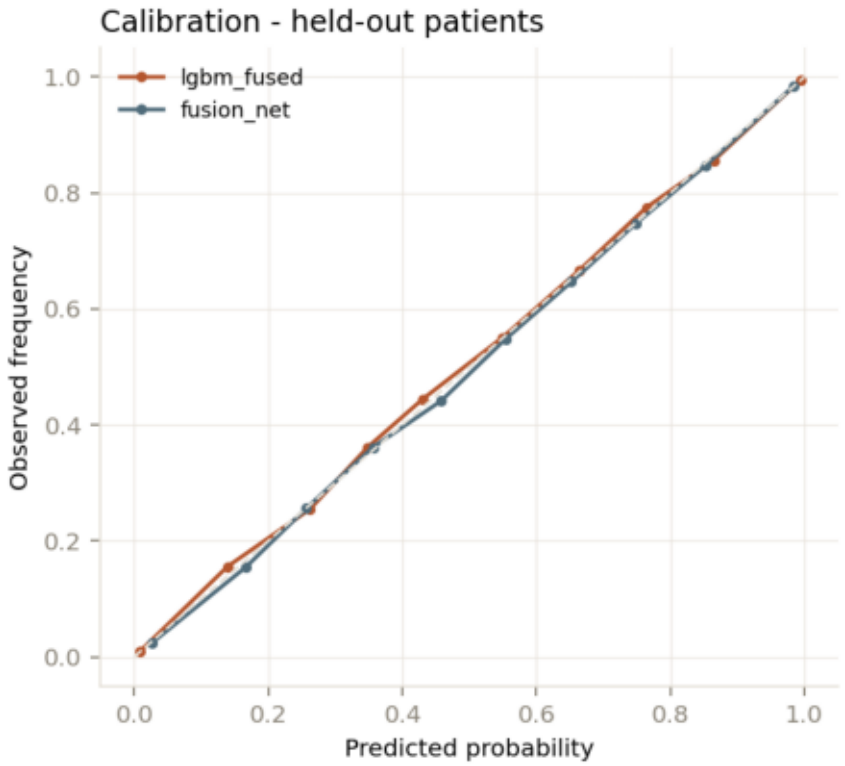
Brier 0.028. 0.45 false alarms per patient-day. Dense 15-minute run: 81% sensitivity, 60 min mean lead.

Every input helps. The sensor helps most.



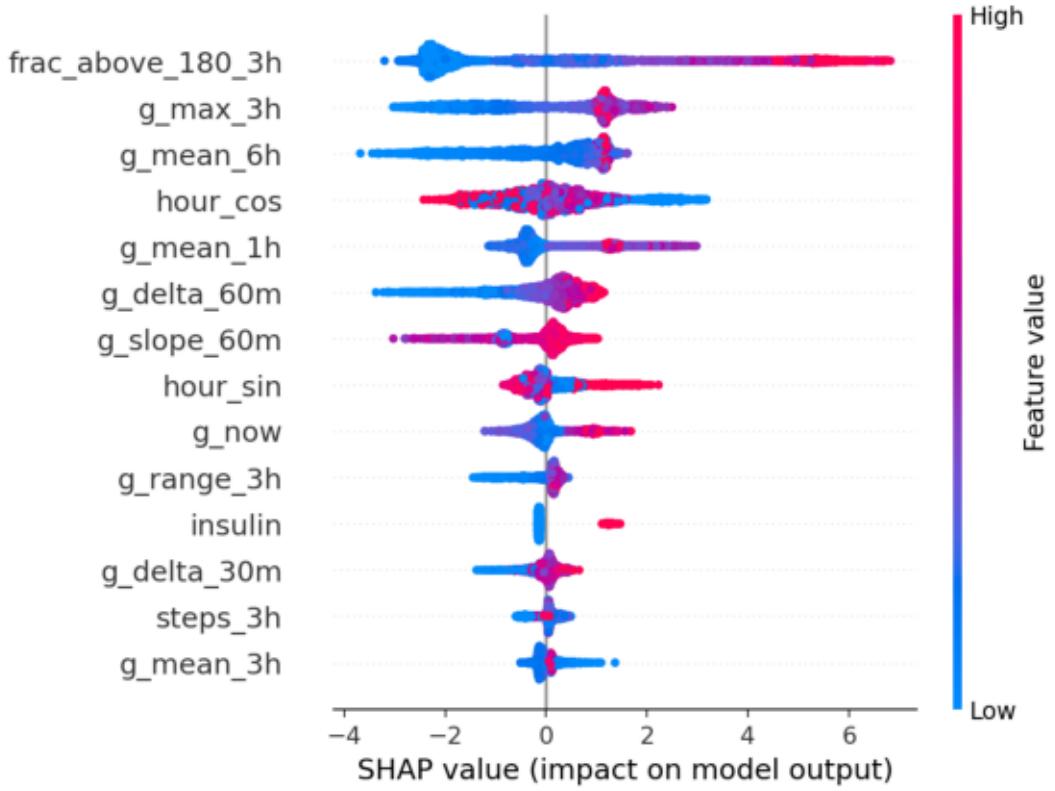
Same checkpoints for every model. EHR-only cannot rank moments inside one patient's day, so its within-patient AUROC is 0.5.

A probability you can act on, not just a ranking.



Brier 0.028 after isotonic calibration. Forecast RMSE 11 / 14 / 23 mg/dL at 30 / 60 / 120 minutes.

Why this alert, in words a clinician can use.

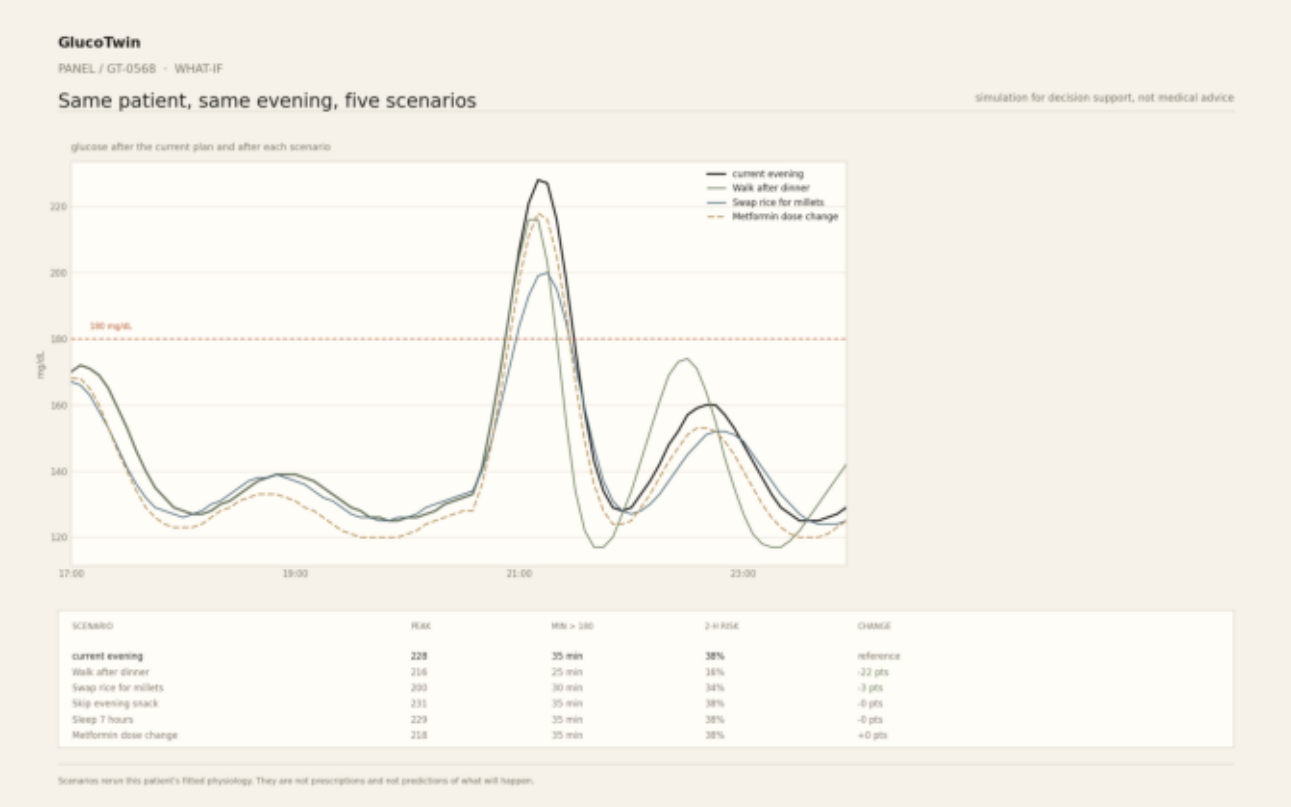


Recent time above range and the size of the current excursion dominate. The record explains who is at risk, not when.

The twin fits the patient, not the population.

- Insulin sensitivity and basal glucose are fitted from the CGM signal.
- Held-out check: R2 0.82, median error 8.4%, Spearman 0.91.
- Those numbers validate the fitting procedure, not the physiology.
- The twin updates as new readings arrive, and keeps its own risk trace.

A walk after dinner cuts the peak by 12 mg/dL.



Same patient, same evening. Risk falls from 38% to 16%. Swapping rice for millets cuts the peak by 28 mg/dL. Skipping the snack does almost nothing.

Built for a clinic, not for a demo.





Cohort sorted by two-hour risk. Twin workspace with forecast band, gauge, time in range, wearables and the alert timeline.

What we did not prove.

- This is synthetic data. No clinical accuracy is claimed.
- External validation is written but not run.
- About 2.7 alerts per patient-day at the deployed threshold.
- Subgroup fairness is not measured.
- It is not a medical device, and it is not approved.

DPDP Act and HIPAA by design. Human in the loop. No automatic treatment change.

What comes next.

- A prospective 8 to 12 week study in one clinic, threshold frozen in advance.
- ABDM and ABHA records, Apple Health and Google Fit, CGM vendor APIs.
- Federated learning across clinics, so records never move.
- A fairer cohort: CKD, SGLT2 and GLP-1, and well-controlled patients.

Thank you.

GlucoTwin is a decision aid, not a doctor. The clinician decides.

Demo video: https://youtu.be/K4raAnGqGx0 · Prototype: https://glucotwin.freebuff.app/